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GITHUB COPILOT SDK

Building an AI Issue Triage Tool with the GitHub Copilot SDK

From empty file to streaming web app in 60 minutes

By Renee Noble ·

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What is the Copilot SDK?

Editor Copilot vs the SDK — a side-by-side comparison

Copilot in your editor

A tool for developers

- Built into VS Code, JetBrains, etc.
- Suggests code while you type
- You see the results on screen

The Copilot SDK

A building block for applications

- A Python / JS / C# library you install
- Your app calls it like any other library
- Your code receives results and decides next steps
- Three parts: Client → Session → Tools

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The Agentic Loop

Prompt → tool call → result → model re-evaluates

STEP 1

Prompt sent

1

"Analyse this issue"

STEP 2

Tool requested

2

e.g. get_github_issue

STEP 3

Tool executes

3

Your code returns data

STEP 4

Result fed back

4

Model reads and decides

STEP 5

Loop or finish

5

More tools if needed

STEP 6

Final response

6

Structured assessment

We don't script the sequence — the model picks tools based on what it needs.

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What we built (Architecture)

One file, four layers, SDK in the middle

① Browser

Pre-built frontend

- EventSource receives SSE
- Renders chat bubbles in real-time
- Lives in src/static/

② FastAPI Server

app.py

- /api/analyse + /api/post-analysis
- Async queue bridges SDK → SSE
- LANG=ja switches to Japanese mode

③ Copilot SDK + Model

gpt-4.1

- Client / Session / Tools
- Generates responses + tool calls
- Streams events back

④ GitHub REST API

Data layer

- Issues / Contents / Code Search
- Auth via GITHUB_TOKEN
- Writes back PostComment / AddLabels

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Next Actions

Clone the repo and try it on your own issues

Workshop course (original)

github.com/reneenoble/github-copilot-sdk-for-beginners

Today's repo (JA branch)

github.com/ChibaYuki347/copilot-sdk-github-issue-analyser/tree/ja

Copilot SDK docs

github.com/github/copilot-sdk

Run command

`python app.py serve`

Star the repo to help others find it ☆